

COLIN VAUGHN

Meridian, ID 83646



colin-v Vaughn.github.io/portfolio



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SUMMARY

3RD YEAR Honors Mechanical and Nuclear Engineering student with experience in R&D bioengineering, robotics, and aerospace systems research. Proficient in CAD, instrumentation, and project coordination. Interested in energy systems, automation, and advanced mechanical design.

EDUCATION

Honors B.S. MECHANICAL & NUCLEAR ENGINEERING

Minors in Applied Mathematics, Mathematics and Physics

Idaho State University (ISU) College of Science and Engineering
Pocatello, ID | Expected May 2028

Accolades: 3.84 GPA, Dean's List (2024- present), University Honors Program (2024) ISU Presidential Scholarship (2024)

SKILLS & COMPETENCES

Engineering Methods: Agile Development, Bill of Materials Management, Technical Documentation, Design-for-Manufacturing, Microsoft Office Suite

Mechanical Design: SolidWorks, AutoCAD, Mechanical Modeling, Rapid Prototyping

Programming & Computing: HMI Development, MATLAB, C++, Python, HTML/CSS

Controls & Robotics: PID, Feedback-driven Autonomous Systems, Mechatronics

Design & Visualization: Adobe Photoshop, Adobe Illustrator, Adobe Premiere Pro

PROJECT HIGHLIGHTS

- Reel Reactor® R&D phase assembly, programming, and evaluation
- Turbine compression chamber analysis via CAD, prototyping, and coded simulations
- Radionuclide interactive decay chain modeling program in Python
- VEX Robotics PID drive control system & autonomous code

EXPERIENCE

May 2026 – MECHANICAL ENGINEERING INTERN

Aug 2026 Reel Reactor Inc. | Star, ID

- **Mechatronics & Programming:** Programmed a Human-Machine Interface and Python data bridge to enable manual control and runtime autonomous operation of an R&D bioreactor
- **System Optimization:** Participated in rapid prototyping, design-for-manufacturing workflows, and technical reviews with senior engineers to evaluate and improve designs, maintain a bill of materials
- **Business Operations:** Presented technical milestones in investor meetings to support ongoing project fundraising

Feb 2026 – AEROSPACE SYSTEMS RESEARCH LAB MEMBER

Present Measurement & Control Engineering Research Center | Pocatello, ID

- **Machine Design:** Designing and fabricating testing interfaces to simulate pressure accumulation in a jet turbine and evaluate optimized compression chamber geometry
- **Instrumentation Integration:** Incorporated systems for iterative data collection and performance evaluation
- **Project Coordination:** Synchronized efforts in a multidisciplinary team of undergraduates, graduates and faculty

Aug 2026 - MATH & PHYSICS CONTENT TUTOR

Present Idaho State University Tutoring Center | Pocatello, ID

- **Content Mastery:** Deepened working knowledge of math and physics concepts via collaborative problem-solving
- **Concept Communication:** Applied various teaching strategies to communicate complex engineering course concepts through physical demonstrations and verbal explanations

Oct 2021 – COMPETITIVE ROBOTICS TEAM CAPTAIN

Apr 2024 VEX Robotics Competition Team | Meridian, ID

- **Agile Project Development:** Directed iterative subsystem refinement involving sensors, manipulators, drivetrains, and PID controls for regional and international team competitions
- **Technical Documentation:** Maintained a multi-volume, award-winning engineering notebook documenting CAD models, design iterations, and project budgeting
- **Team Leadership:** Led an 8-member team in full development cycles, allocated tasks, and fostered collaboration